

EDA — Motor Insurance Data 2011–2018

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2026-06-09

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```

library(readr)
library(dplyr)
library(ggplot2)
library(tidyr)
library(scales)
library(forcats)      # fct_lump, fct_reorder
library(lubridate)    # year(), month()
library(patchwork)    # menyusun beberapa plot menjadi satu panel
library(GGally)       # ggpairs untuk scatter matrix
library(corrplot)     # correlation heatmap

theme_set(theme_minimal(base_size = 12) +
  theme(plot.title      = element_text(face = "bold", size = 13),
        plot.subtitle   = element_text(colour = "grey40"),
        axis.title      = element_text(size = 11)))

motor <- read_csv("vehicle_insurance_data/motor_data_clean.csv") %>%
  mutate(
    SEX          = as.factor(SEX),
    INSR_TYPE    = as.factor(INSR_TYPE),
    TYPE_VEHICLE = as.factor(TYPE_VEHICLE),
    MAKE         = as.factor(MAKE),
    USAGE        = as.factor(USAGE),
    CLAIM_IND    = as.factor(CLAIM_IND),
    COVERAGE_TYPE = as.factor(COVERAGE_TYPE),
    INSR_BEGIN   = as.Date(INSR_BEGIN),
    INSR_END     = as.Date(INSR_END)
  )

cat("Dimensi dataset bersih:", nrow(motor), "baris x", ncol(motor), "kolom\n")

## Dimensi dataset bersih: 801648 baris x 18 kolom

```

BAGIAN 1 — VARIABEL RESPONS

1.1 Distribusi CLAIM_IND (Indikator Klaim Biner)

```
# Ringkasan frekuensi
tbl_claim <- motor %>%
  count(CLAIM_IND) %>%
  mutate(
    label = ifelse(CLAIM_IND == 1, "Klaim (1)", "Tidak Klaim (0)"),
    pct = n / sum(n),
    pct_txt = percent(pct, accuracy = 0.01)
  )

print(tbl_claim)

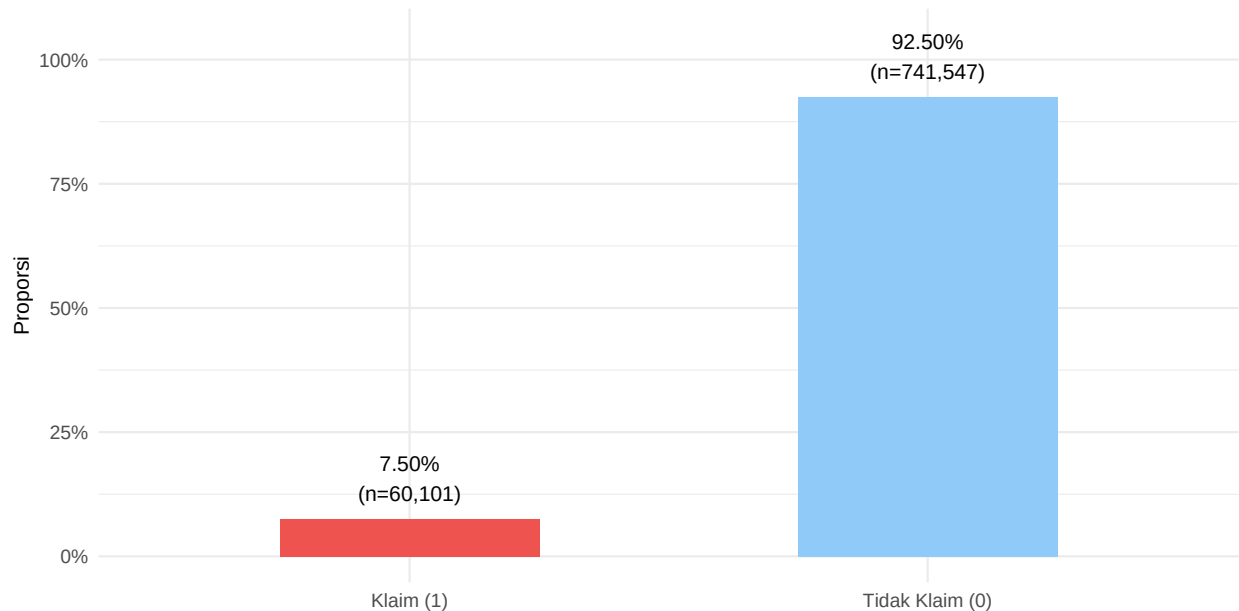
## # A tibble: 2 x 5
##   CLAIM_IND      n label          pct pct_txt
##   <fct>      <int> <chr>          <dbl> <chr>
## 1 0        741547 Tidak Klaim (0) 0.925  92.5%
## 2 1         60101 Klaim (1)      0.075   7.5%

# Bar chart dengan proporsi
p_claim_bar <- ggplot(tbl_claim, aes(x = label, y = pct, fill = CLAIM_IND)) +
  geom_col(width = 0.5, show.legend = FALSE) +
  geom_text(aes(label = paste0(pct_txt, "\nn(n=", comma(n), ")")),
    vjust = -0.4, size = 3.8) +
  scale_y_continuous(labels = percent_format(), limits = c(0, 1.05)) +
  scale_fill_manual(values = c("0" = "#90CAF9", "1" = "#EF5350")) +
  labs(title = "Gambar 1. Distribusi CLAIM_IND",
    subtitle = "Dataset bersifat strongly zero-inflated (~7,5% klaim)",
    x = NULL, y = "Proporsi")

p_claim_bar
```

Gambar 1. Distribusi CLAIM_IND

Dataset bersifat strongly zero-inflated (~7,5% klaim)



1.2 Distribusi CLAIM_PAID (hanya polis berklaim)

```
klaim_df <- motor %>% filter(!is.na(CLAIM_PAID) & CLAIM_PAID > 0)
```

```
cat("Statistik deskriptif CLAIM_PAID (polis berklaim):\n")
```

```
## Statistik deskriptif CLAIM_PAID (polis berklaim):
```

```
summary(klaim_df$CLAIM_PAID)
```

```
##      Min.   1st Qu.   Median     Mean   3rd Qu.    Max.
##         5    13518    34700   256652   134059 152445765
```

```
# Histogram skala asli
```

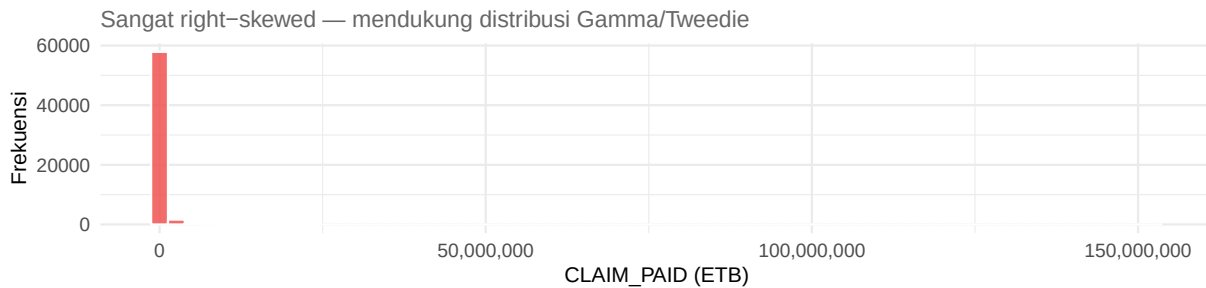
```
p_cp_hist <- ggplot(klaim_df, aes(x = CLAIM_PAID)) +
  geom_histogram(bins = 60, fill = "#EF5350", colour = "white", alpha = 0.85) +
  scale_x_continuous(labels = comma) +
  labs(title = "Gambar 2a. Distribusi CLAIM_PAID (skala asli)",
       subtitle = "Sangat right-skewed - mendukung distribusi Gamma/Tweedie",
       x = "CLAIM_PAID (ETB)", y = "Frekuensi")
```

```
# Histogram log-skala
```

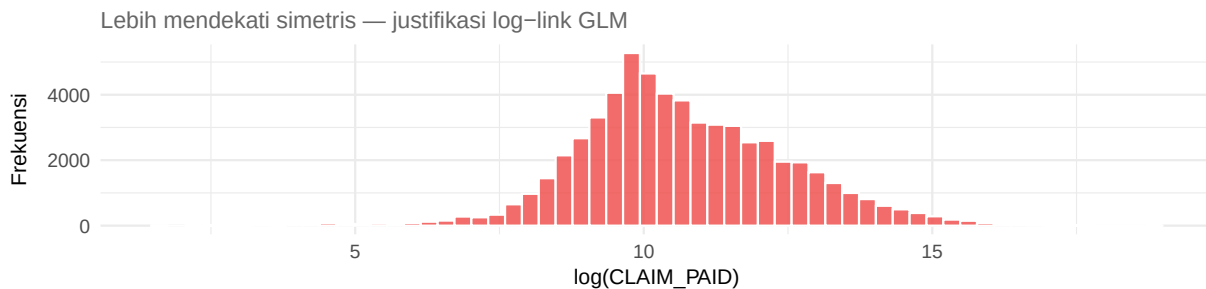
```
p_cp_log <- ggplot(klaim_df, aes(x = log(CLAIM_PAID))) +
  geom_histogram(bins = 60, fill = "#EF5350", colour = "white", alpha = 0.85) +
  labs(title = "Gambar 2b. Distribusi log(CLAIM_PAID)",
       subtitle = "Lebih mendekati simetris - justifikasi log-link GLM",
       x = "log(CLAIM_PAID)", y = "Frekuensi")
```

```
p_cp_hist / p_cp_log # patchwork: susun vertikal
```

Gambar 2a. Distribusi CLAIM_PAID (skala asli)



Gambar 2b. Distribusi log(CLAIM_PAID)



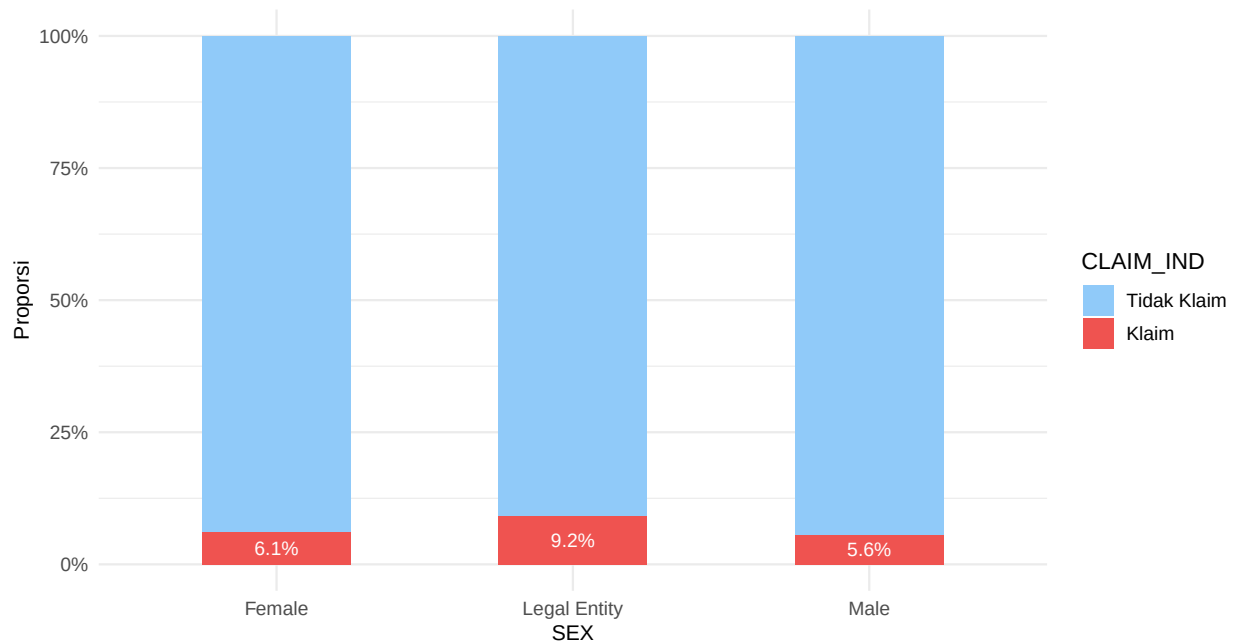
BAGIAN 2 — VARIABEL KATEGORIS

2.1 SEX

```
p_sex_bar <- motor %>%
  count(SEX, CLAIM_IND) %>%
  group_by(SEX) %>%
  mutate(pct = n / sum(n)) %>%
  ungroup() %>%
  ggplot(aes(x = SEX, y = pct, fill = CLAIM_IND)) +
  geom_col(position = "fill", width = 0.5) +
  geom_text(aes(label = ifelse(CLAIM_IND == 1, percent(pct, accuracy = 0.1), ""),
    position = position_fill(vjust = 0.5), size = 3.5, colour = "white") +
  scale_y_continuous(labels = percent_format()) +
  scale_fill_manual(values = c("0" = "#90CAF9", "1" = "#EF5350"),
    labels = c("Tidak Klaim", "Klaim")) +
  labs(title = "Gambar 3. Proporsi Klaim per Jenis Kelamin / Entitas",
    x = "SEX", y = "Proporsi", fill = "CLAIM_IND")

p_sex_bar
```

Gambar 3. Proporsi Klaim per Jenis Kelamin / Entitas

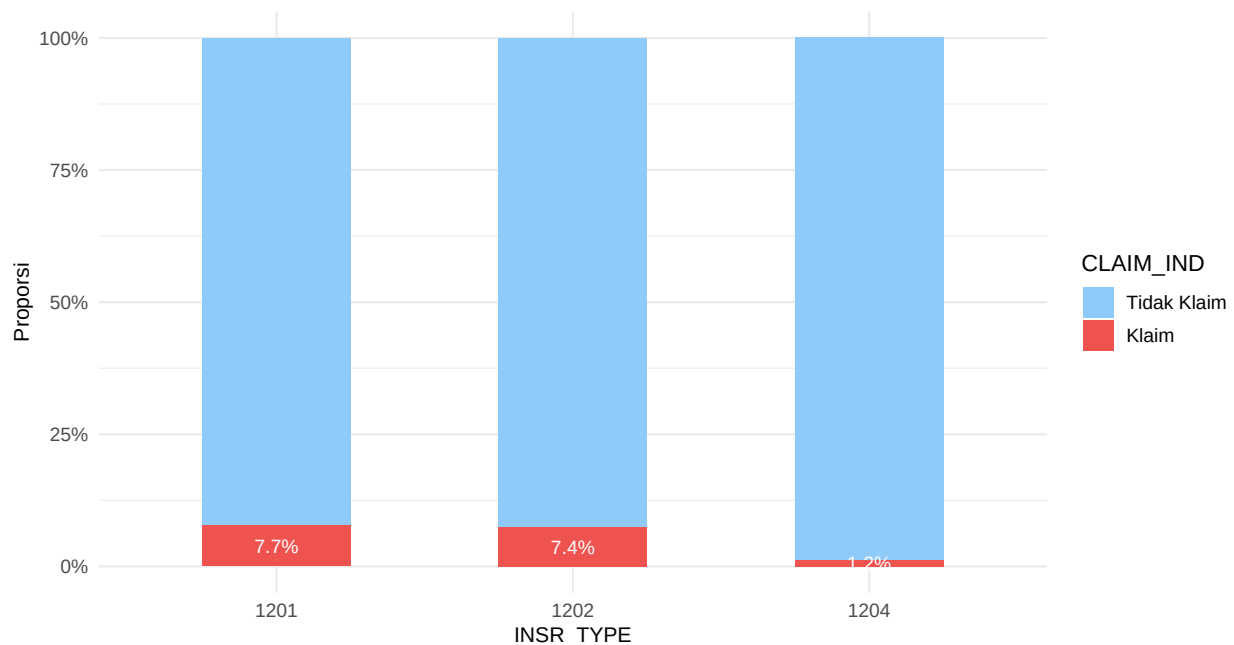


2.2 INSR_TYPE

```
p_insr_bar <- motor %>%
  count(INSR_TYPE, CLAIM_IND) %>%
  group_by(INSR_TYPE) %>%
  mutate(pct = n / sum(n)) %>%
  ungroup() %>%
  ggplot(aes(x = INSR_TYPE, y = pct, fill = CLAIM_IND)) +
  geom_col(position = "fill", width = 0.5) +
  geom_text(aes(label = ifelse(CLAIM_IND == 1, percent(pct, accuracy = 0.1), ""),
    position = position_fill(vjust = 0.5), size = 3.5, colour = "white")) +
  scale_y_continuous(labels = percent_format()) +
  scale_fill_manual(values = c("0" = "#90CAF9", "1" = "#EF5350"),
    labels = c("Tidak Klaim", "Klaim")) +
  labs(title = "Gambar 4. Proporsi Klaim per Tipe Asuransi",
    x = "INSR_TYPE", y = "Proporsi", fill = "CLAIM_IND")
```

p_insr_bar

Gambar 4. Proporsi Klaim per Tipe Asuransi



2.3 TYPE_VEHICLE

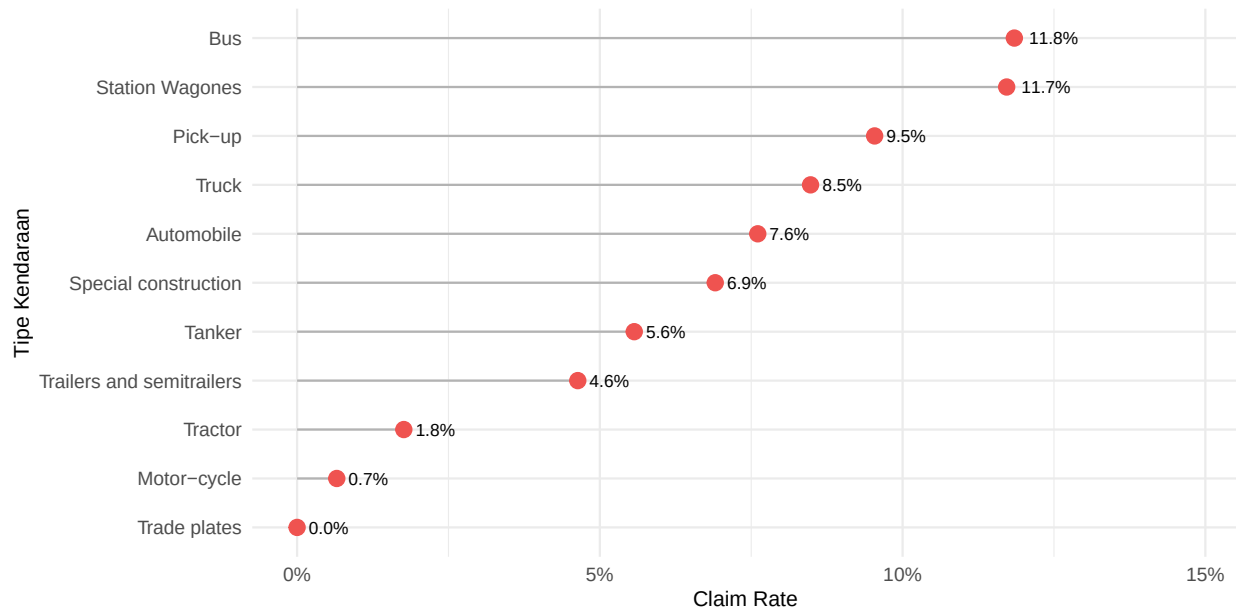
```
# Hitung claim rate per tipe kendaraan
claim_rate_vehicle <- motor %>%
  group_by(TYPE_VEHICLE) %>%
  summarise(
    n_polis      = n(),
    n_klaim      = sum(CLAIM_IND == 1),
    claim_rate   = n_klaim / n_polis
  ) %>%
  arrange(desc(claim_rate))

# Lollipop chart claim rate
p_vehicle_lollipop <- ggplot(claim_rate_vehicle,
  aes(x = claim_rate,
    y = fct_reorder(TYPE_VEHICLE, claim_rate))) +
  geom_segment(aes(x = 0, xend = claim_rate,
    y = fct_reorder(TYPE_VEHICLE, claim_rate),
    yend = fct_reorder(TYPE_VEHICLE, claim_rate)),
    colour = "grey70") +
  geom_point(colour = "#EF5350", size = 3.5) +
  geom_text(aes(label = percent(claim_rate, accuracy = 0.1)),
    hjust = -0.3, size = 3.2) +
  scale_x_continuous(labels = percent_format(), limits = c(0, max(claim_rate_vehicle$claim_rate) * 1.25),
  labs(title = "Gambar 5. Claim Rate per Tipe Kendaraan",
    subtitle = "Klaim rate = proporsi polis yang mengajukan klaim",
    x = "Claim Rate", y = "Tipe Kendaraan")

p_vehicle_lollipop
```

Gambar 5. Claim Rate per Tipe Kendaraan

Klaim rate = proporsi polis yang mengajukan klaim



2.4 USAGE

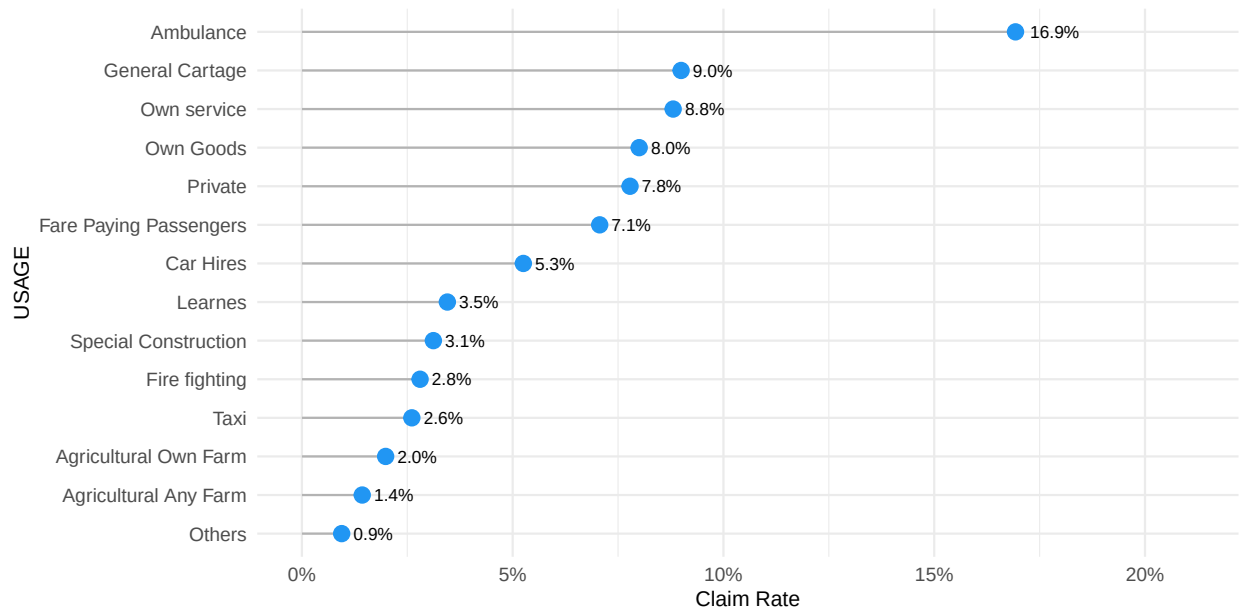
```
claim_rate_usage <- motor %>%
  group_by(USAGE) %>%
  summarise(
    n_polis    = n(),
    n_klaim    = sum(CLAIM_IND == 1),
    claim_rate = n_klaim / n_polis
  ) %>%
  arrange(desc(claim_rate))

p_usage_lollipop <- ggplot(claim_rate_usage,
  aes(x = claim_rate,
      y = fct_reorder(USAGE, claim_rate))) +
  geom_segment(aes(x = 0, xend = claim_rate,
                  y = fct_reorder(USAGE, claim_rate),
                  yend = fct_reorder(USAGE, claim_rate)),
    colour = "grey70") +
  geom_point(colour = "#2196F3", size = 3.5) +
  geom_text(aes(label = percent(claim_rate, accuracy = 0.1)),
    hjust = -0.3, size = 3.2) +
  scale_x_continuous(labels = percent_format(),
    limits = c(0, max(claim_rate_usage$claim_rate) * 1.25)) +
  labs(title = "Gambar 6. Claim Rate per Penggunaan Kendaraan",
    subtitle = "Kategori USAGE dengan klaim rate tertinggi menjadi kandidat prediktor kuat",
    x = "Claim Rate", y = "USAGE")

p_usage_lollipop
```

Gambar 6. Claim Rate per Penggunaan Kendaraan

Kategori USAGE dengan klaim rate tertinggi menjadi kandidat prediktor kuat



2.5 MAKE (Top 20 Merek)

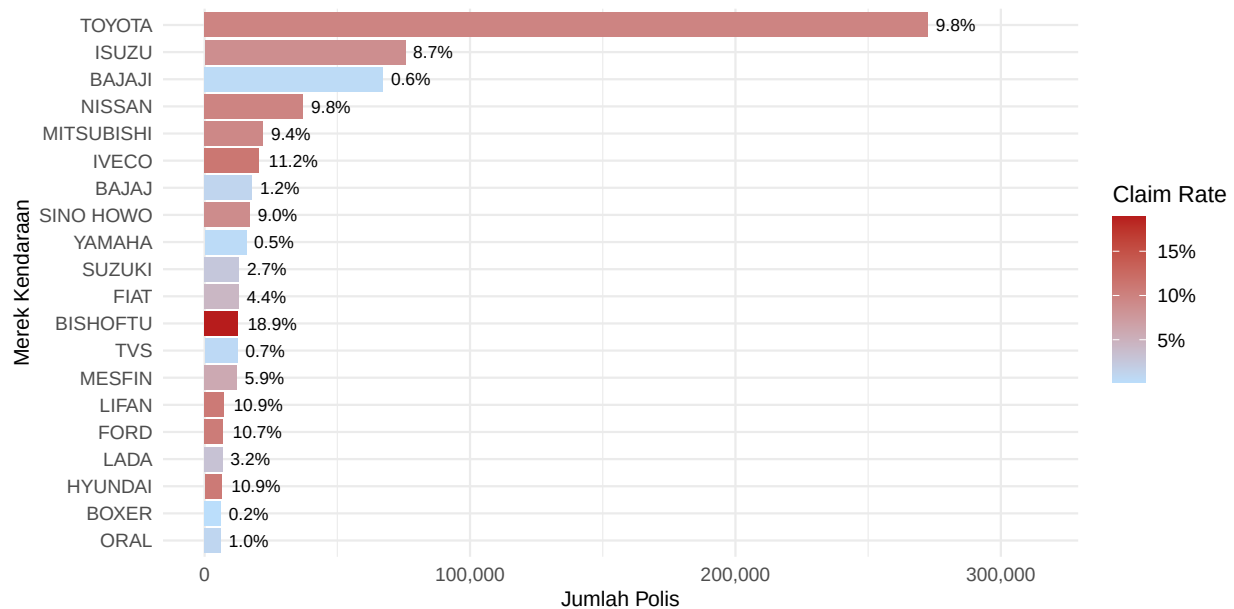
```
# MAKE memiliki 792 level - hanya tampilkan top 20 berdasarkan volume
top20_make <- motor %>%
  group_by(MAKE) %>%
  summarise(
    n_polis = n(),
    n_klaim = sum(CLAIM_IND == 1),
    claim_rate = n_klaim / n_polis
  ) %>%
  arrange(desc(n_polis)) %>%
  slice_head(n = 20)

p_make <- ggplot(top20_make,
  aes(x = n_polis,
    y = fct_reorder(MAKE, n_polis),
    fill = claim_rate)) +
  geom_col() +
  geom_text(aes(label = percent(claim_rate, accuracy = 0.1)),
    hjust = -0.2, size = 3) +
  scale_x_continuous(labels = comma,
    limits = c(0, max(top20_make$n_polis) * 1.15)) +
  scale_fill_gradient(low = "#BBDEFB", high = "#B71C1C",
    labels = percent_format(),
    name = "Claim Rate") +
  labs(title = "Gambar 7. Top 20 Merek Kendaraan (Volume Polis)",
    subtitle = "Warna menunjukkan claim rate; teks menunjukkan nilai claim rate",
    x = "Jumlah Polis", y = "Merek Kendaraan")

p_make
```

Gambar 7. Top 20 Merek Kendaraan (Volume Polis)

Warna menunjukkan claim rate; teks menunjukkan nilai claim rate

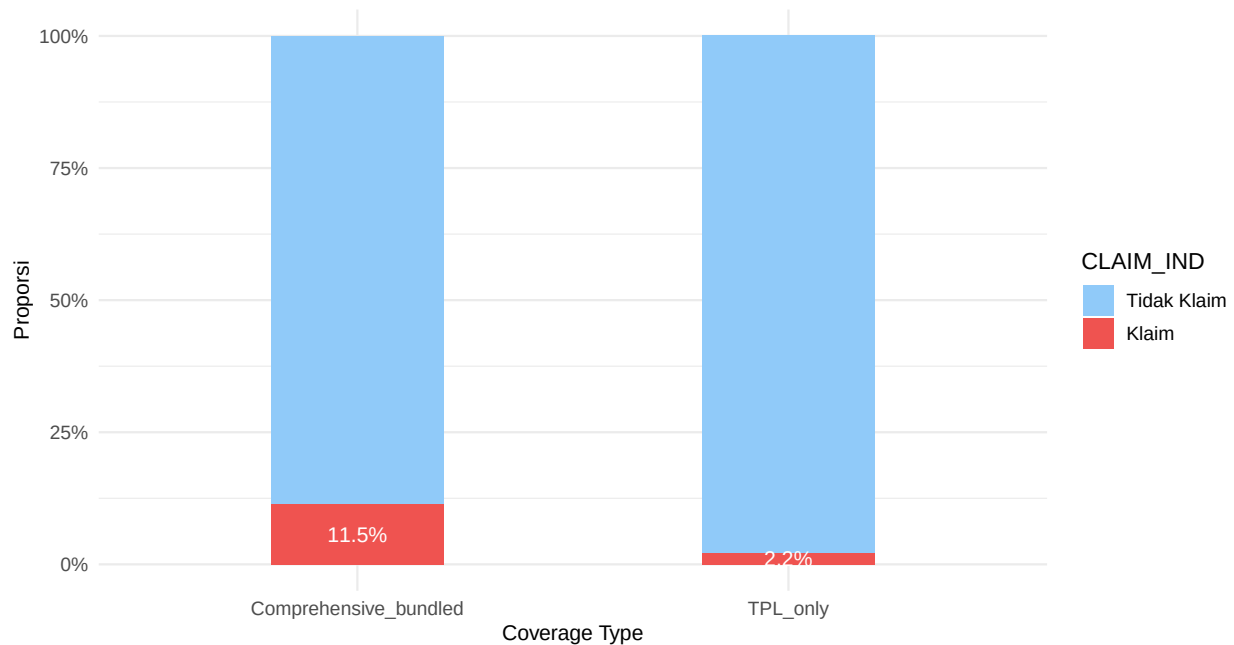


2.6 COVERAGE_TYPE

```
p_coverage <- motor %>%
  count(COVERAGE_TYPE, CLAIM_IND) %>%
  group_by(COVERAGE_TYPE) %>%
  mutate(pct = n / sum(n)) %>%
  ungroup() %>%
  ggplot(aes(x = COVERAGE_TYPE, y = pct, fill = CLAIM_IND)) +
  geom_col(position = "fill", width = 0.4) +
  geom_text(aes(label = ifelse(CLAIM_IND == 1, percent(pct, accuracy = 0.1), ""),
    position = position_fill(vjust = 0.5), size = 3.8, colour = "white")) +
  scale_y_continuous(labels = percent_format()) +
  scale_fill_manual(values = c("0" = "#90CAF9", "1" = "#EF5350"),
    labels = c("Tidak Klaim", "Klaim")) +
  labs(title = "Gambar 8. Proporsi Klaim per Tipe Coverage",
    x = "Coverage Type", y = "Proporsi", fill = "CLAIM_IND")
```

p_coverage

Gambar 8. Proporsi Klaim per Tipe Coverage



BAGIAN 3 — VARIABEL NUMERIK KONTINU

3.1 Ringkasan Statistik Variabel Numerik

```
num_vars <- motor %>%
  select(INSURED_VALUE, PREMIUM, PROD_YEAR, CCM_TON,
         CLAIM_PAID, DURATION_DAYS, EXPOSURE_YR)

cat("Statistik deskriptif variabel numerik:\n")
```

```
## Statistik deskriptif variabel numerik:
```

```
summary(num_vars)
```

```
##  INSURED_VALUE      PREMIUM      PROD_YEAR      CCM_TON
##  Min.   :      0  Min.   :      0  Min.   :1950  Min.   :      0
##  1st Qu.:      0  1st Qu.:   757  1st Qu.:2000  1st Qu.:   200
##  Median : 180000  Median :   3379  Median :2008  Median :  2494
##  Mean   : 509203  Mean   :   7409  Mean   :2005  Mean   :  3173
##  3rd Qu.: 730000  3rd Qu.:   9645  3rd Qu.:2012  3rd Qu.:  4164
##  Max.   :250000000  Max.   :7581230  Max.   :2018  Max.   : 20000
##
##    CLAIM_PAID      DURATION_DAYS      EXPOSURE_YR
##  Min.   :      0  Min.   :  1.0  Min.   :0.002738
##  1st Qu.: 13500  1st Qu.:364.0  1st Qu.:0.996578
##  Median : 34633  Median :364.0  Median :0.996578
##  Mean   : 256477  Mean   :351.4  Mean   :0.962205
```

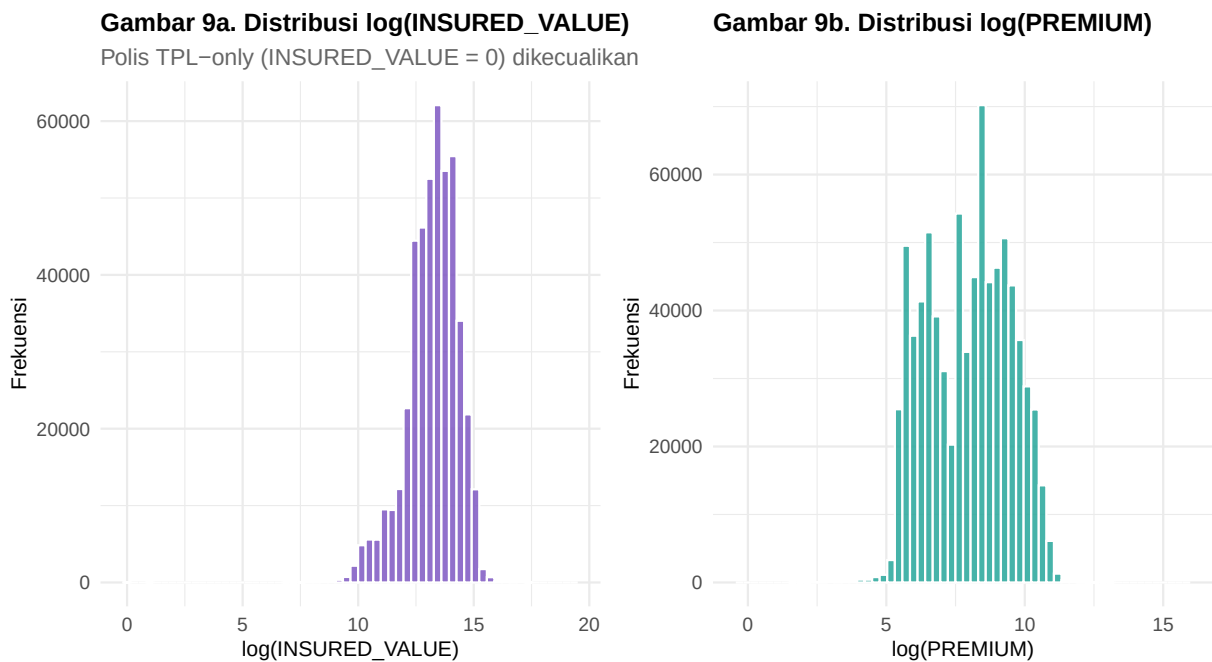
```
## 3rd Qu.: 133846 3rd Qu.:364.0 3rd Qu.:0.996578
## Max. :152445765 Max. :523.0 Max. :1.431896
## NA's :741506
```

3.2 Distribusi INSURED_VALUE dan PREMIUM

```
# Histogram INSURED_VALUE - termasuk polis TPL (INSURED_VALUE = 0)
p_iv <- ggplot(motor %>% filter(INSURED_VALUE > 0),
              aes(x = log(INSURED_VALUE))) +
  geom_histogram(bins = 60, fill = "#7E57C2", colour = "white", alpha = 0.85) +
  labs(title = "Gambar 9a. Distribusi log(INSURED_VALUE)",
       subtitle = "Polis TPL-only (INSURED_VALUE = 0) dikecualikan",
       x = "log(INSURED_VALUE)", y = "Frekuensi")

# Histogram PREMIUM
p_prem <- ggplot(motor %>% filter(PREMIUM > 0),
                aes(x = log(PREMIUM))) +
  geom_histogram(bins = 60, fill = "#26A69A", colour = "white", alpha = 0.85) +
  labs(title = "Gambar 9b. Distribusi log(PREMIUM)",
       x = "log(PREMIUM)", y = "Frekuensi")

p_iv | p_prem # patchwork: susun horizontal
```



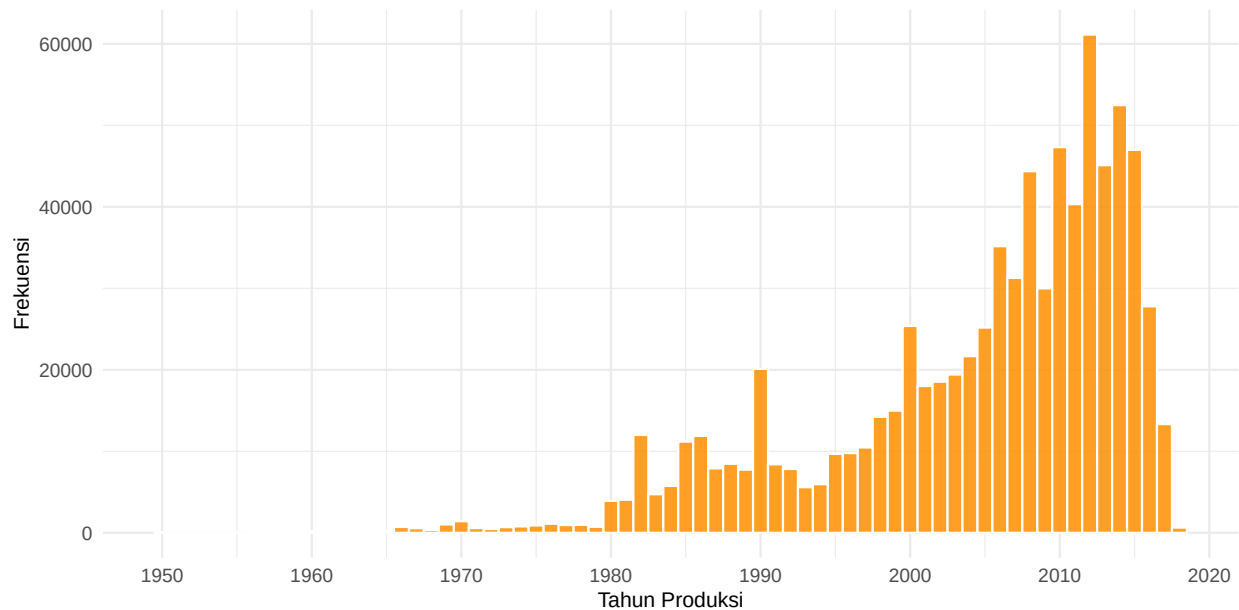
3.3 Distribusi PROD_YEAR (Tahun Produksi Kendaraan)

```
p_prod <- ggplot(motor, aes(x = PROD_YEAR)) +
  geom_histogram(binwidth = 1, fill = "#FF8F00", colour = "white", alpha = 0.85) +
  scale_x_continuous(breaks = seq(1950, 2020, by = 10)) +
  labs(title = "Gambar 10. Distribusi Tahun Produksi Kendaraan",
       subtitle = "Mayoritas kendaraan diproduksi setelah tahun 1995",
       x = "Tahun Produksi", y = "Frekuensi")
```

p_prod

Gambar 10. Distribusi Tahun Produksi Kendaraan

Mayoritas kendaraan diproduksi setelah tahun 1995



3.4 VEHICLE_AGE (Usia Kendaraan) — Derived Variable

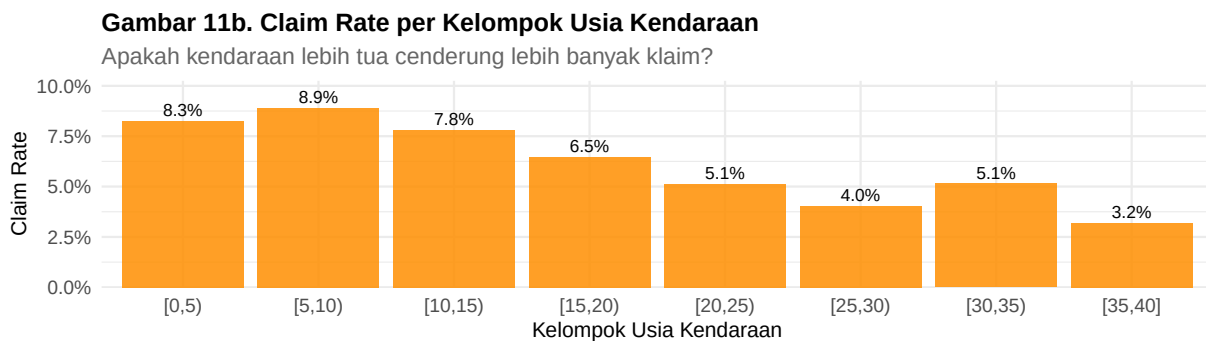
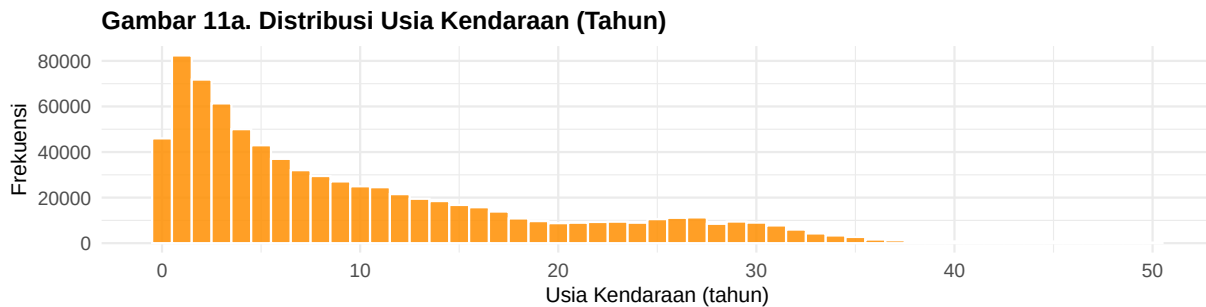
```
# Membuat variabel usia kendaraan dari INSR_BEGIN dan PROD_YEAR
motor <- motor %>%
  mutate(VEHICLE_AGE = year(INSR_BEGIN) - PROD_YEAR)

# Distribusi usia kendaraan
p_vage_hist <- ggplot(motor %>% filter(VEHICLE_AGE >= 0 & VEHICLE_AGE <= 50),
  aes(x = VEHICLE_AGE)) +
  geom_histogram(binwidth = 1, fill = "#FF8F00", colour = "white", alpha = 0.85) +
  labs(title = "Gambar 11a. Distribusi Usia Kendaraan (Tahun)",
    x = "Usia Kendaraan (tahun)", y = "Frekuensi")

# Claim rate per usia kendaraan (kelompok 5 tahun)
p_vage_rate <- motor %>%
  filter(VEHICLE_AGE >= 0 & VEHICLE_AGE <= 40) %>%
  mutate(AGE_GROUP = cut(VEHICLE_AGE, breaks = seq(0, 40, by = 5),
    include.lowest = TRUE, right = FALSE)) %>%
  group_by(AGE_GROUP) %>%
  summarise(
    n_polis = n(),
    claim_rate = mean(CLAIM_IND == 1)
  ) %>%
  ggplot(aes(x = AGE_GROUP, y = claim_rate)) +
  geom_col(fill = "#FF8F00", alpha = 0.85) +
  geom_text(aes(label = percent(claim_rate, accuracy = 0.1)),
    vjust = -0.4, size = 3.2) +
  scale_y_continuous(labels = percent_format(), expand = expansion(mult = c(0, 0.15))) +
```

```
labs(title = "Gambar 11b. Claim Rate per Kelompok Usia Kendaraan",
      subtitle = "Apakah kendaraan lebih tua cenderung lebih banyak klaim?",
      x = "Kelompok Usia Kendaraan", y = "Claim Rate")
```

p_vage_hist / p_vage_rate



3.5 Distribusi DURATION_DAYS dan EXPOSURE_YR

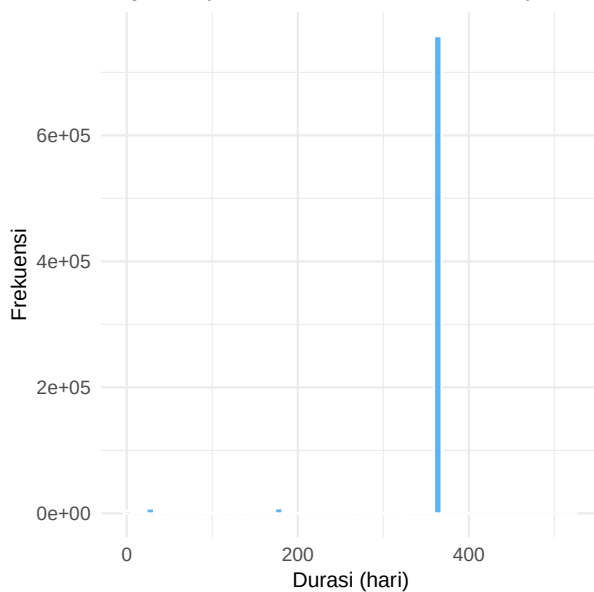
```
p_dur <- ggplot(motor, aes(x = DURATION_DAYS)) +
  geom_histogram(bins = 60, fill = "#42A5F5", colour = "white", alpha = 0.85) +
  labs(title = "Gambar 12a. Distribusi Durasi Polis (hari)",
        subtitle = "Mayoritas polis berdurasi sekitar 1 tahun (365 hari)",
        x = "Durasi (hari)", y = "Frekuensi")
```

```
p_exp <- ggplot(motor, aes(x = EXPOSURE_YR)) +
  geom_histogram(bins = 60, fill = "#42A5F5", colour = "white", alpha = 0.85) +
  labs(title = "Gambar 12b. Distribusi Exposure (tahun)",
        x = "Exposure (tahun)", y = "Frekuensi")
```

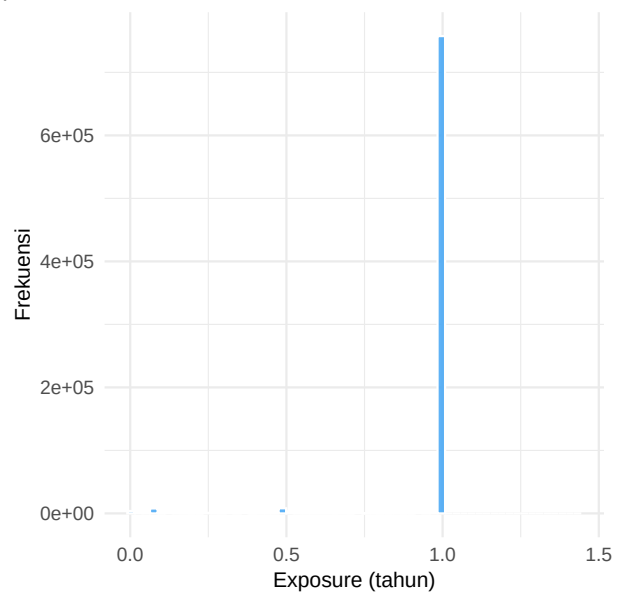
p_dur | p_exp

Gambar 12a. Distribusi Durasi Polis (hari)

Mayoritas polis berdurasi sekitar 1 tahun (365 hari)



Gambar 12b. Distribusi Exposure (tahun)



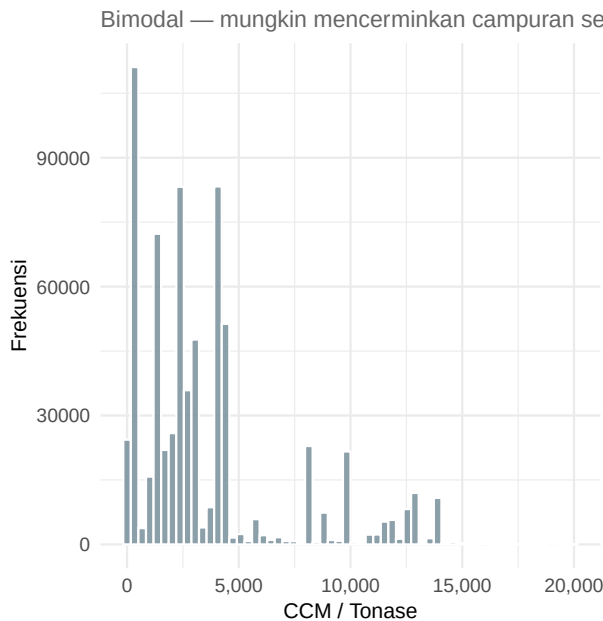
3.6 CCM_TON (Kapasitas Mesin / Tonase)

```
p_ccm <- ggplot(motor %>% filter(CCM_TON > 0),
               aes(x = CCM_TON)) +
  geom_histogram(bins = 60, fill = "#78909C", colour = "white", alpha = 0.85) +
  scale_x_continuous(labels = comma) +
  labs(title = "Gambar 13a. Distribusi CCM_TON (skala asli)",
       subtitle = "Bimodal - mungkin mencerminkan campuran sepeda motor dan kendaraan besar",
       x = "CCM / Tonase", y = "Frekuensi")

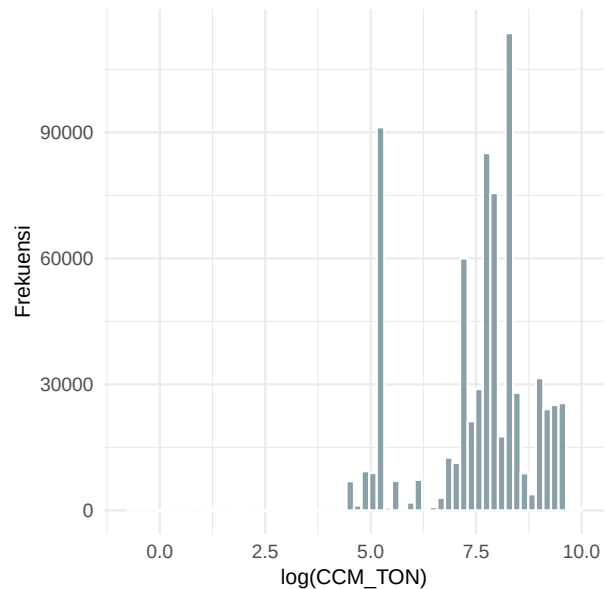
p_ccm_log <- ggplot(motor %>% filter(CCM_TON > 0),
                   aes(x = log(CCM_TON))) +
  geom_histogram(bins = 60, fill = "#78909C", colour = "white", alpha = 0.85) +
  labs(title = "Gambar 13b. Distribusi log(CCM_TON)",
       x = "log(CCM_TON)", y = "Frekuensi")

p_ccm | p_ccm_log
```

Gambar 13a. Distribusi CCM_TON (skala asli)



Gambar 13b. Distribusi log(CCM_TON)



BAGIAN 4 — VARIABEL TANGGAL (TREN TEMPORAL)

4.1 Tren Jumlah Polis per Tahun

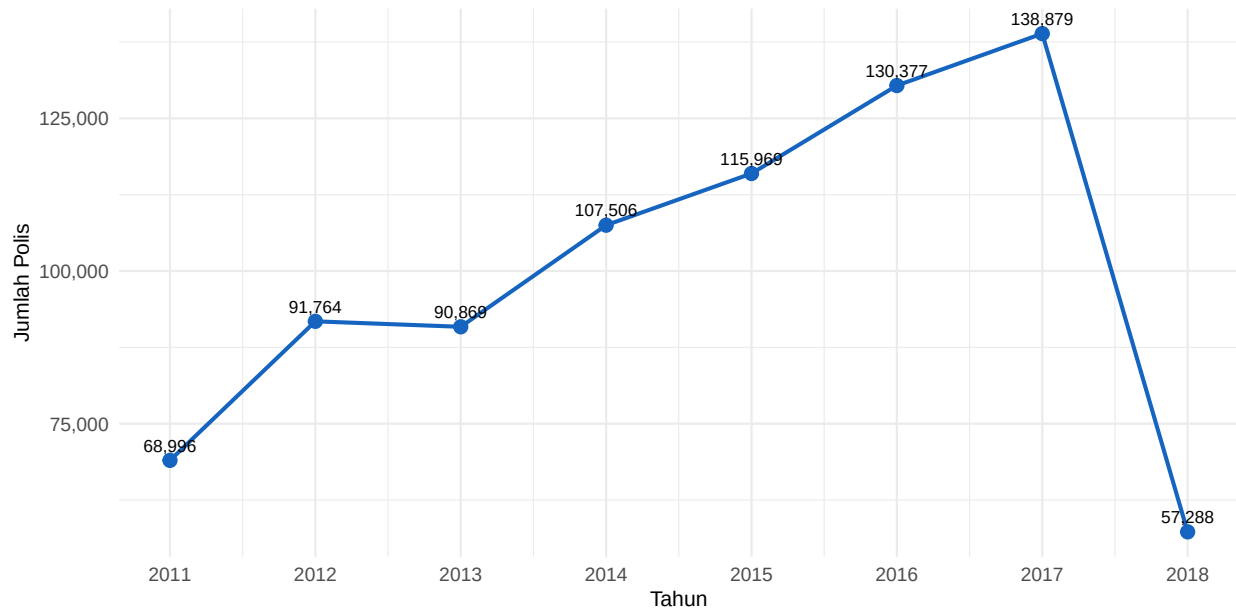
```
motor <- motor %>%
  mutate(
    TAHUN = year(INSR_BEGIN),
    BULAN = month(INSR_BEGIN, label = TRUE, abbr = TRUE)
  )

p_trend_polis <- motor %>%
  count(TAHUN) %>%
  ggplot(aes(x = TAHUN, y = n)) +
  geom_line(colour = "#1565C0", linewidth = 1) +
  geom_point(colour = "#1565C0", size = 3) +
  geom_text(aes(label = comma(n)), vjust = -0.7, size = 3.2) +
  scale_x_continuous(breaks = 2011:2018) +
  scale_y_continuous(labels = comma) +
  labs(title = "Gambar 14. Tren Jumlah Polis per Tahun (2011-2018)",
       subtitle = "Memvisualisasikan pertumbuhan portofolio asuransi",
       x = "Tahun", y = "Jumlah Polis")

p_trend_polis
```

Gambar 14. Tren Jumlah Polis per Tahun (2011–2018)

Memvisualisasikan pertumbuhan portofolio asuransi



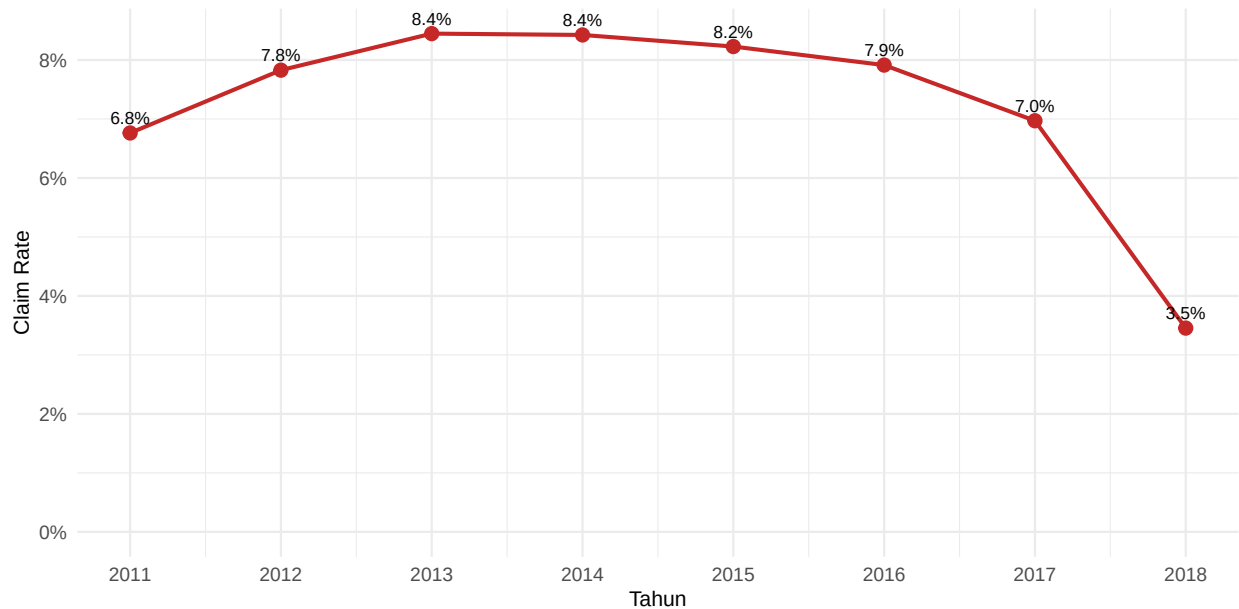
4.2 Tren Claim Rate per Tahun

```
p_trend_rate <- motor %>%
  group_by(TAHUN) %>%
  summarise(
    n_polis      = n(),
    n_klaim      = sum(CLAIM_IND == 1),
    claim_rate   = n_klaim / n_polis
  ) %>%
  ggplot(aes(x = TAHUN, y = claim_rate)) +
  geom_line(colour = "#C62828", linewidth = 1) +
  geom_point(colour = "#C62828", size = 3) +
  geom_text(aes(label = percent(claim_rate, accuracy = 0.1)),
    vjust = -0.7, size = 3.2) +
  scale_x_continuous(breaks = 2011:2018) +
  scale_y_continuous(labels = percent_format(), limits = c(0, NA)) +
  labs(title = "Gambar 15. Tren Claim Rate Tahunan (2011-2018)",
    subtitle = "Apakah claim rate berubah signifikan antar tahun?",
    x = "Tahun", y = "Claim Rate")

p_trend_rate
```

Gambar 15. Tren Claim Rate Tahunan (2011–2018)

Apakah claim rate berubah signifikan antar tahun?



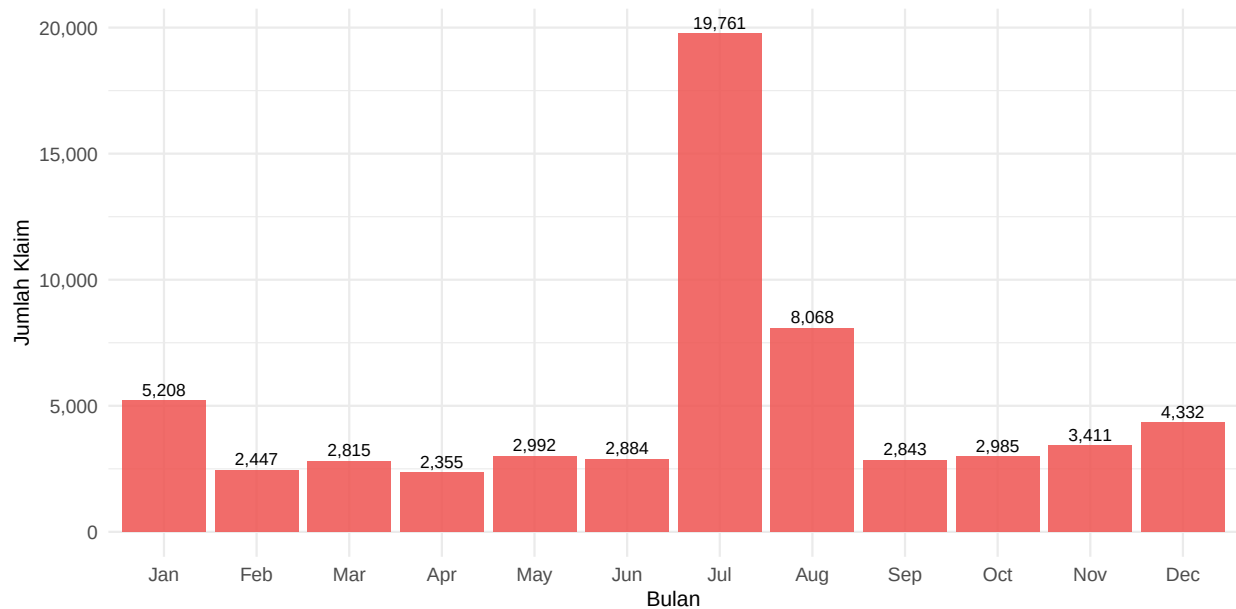
4.3 Pola Musiman — Jumlah Klaim per Bulan

```
p_bulanan <- motor %>%
  filter(CLAIM_IND == 1) %>%
  count(BULAN) %>%
  ggplot(aes(x = BULAN, y = n)) +
  geom_col(fill = "#EF5350", alpha = 0.85) +
  geom_text(aes(label = comma(n)), vjust = -0.4, size = 3.2) +
  scale_y_continuous(labels = comma) +
  labs(title = "Gambar 16. Distribusi Klaim per Bulan (Agregat 2011-2018)",
       subtitle = "Apakah ada pola musiman dalam frekuensi klaim?",
       x = "Bulan", y = "Jumlah Klaim")

p_bulanan
```

Gambar 16. Distribusi Klaim per Bulan (Agregat 2011–2018)

Apakah ada pola musiman dalam frekuensi klaim?



BAGIAN 5 — HUBUNGAN PREDIKTOR DENGAN RESPONS

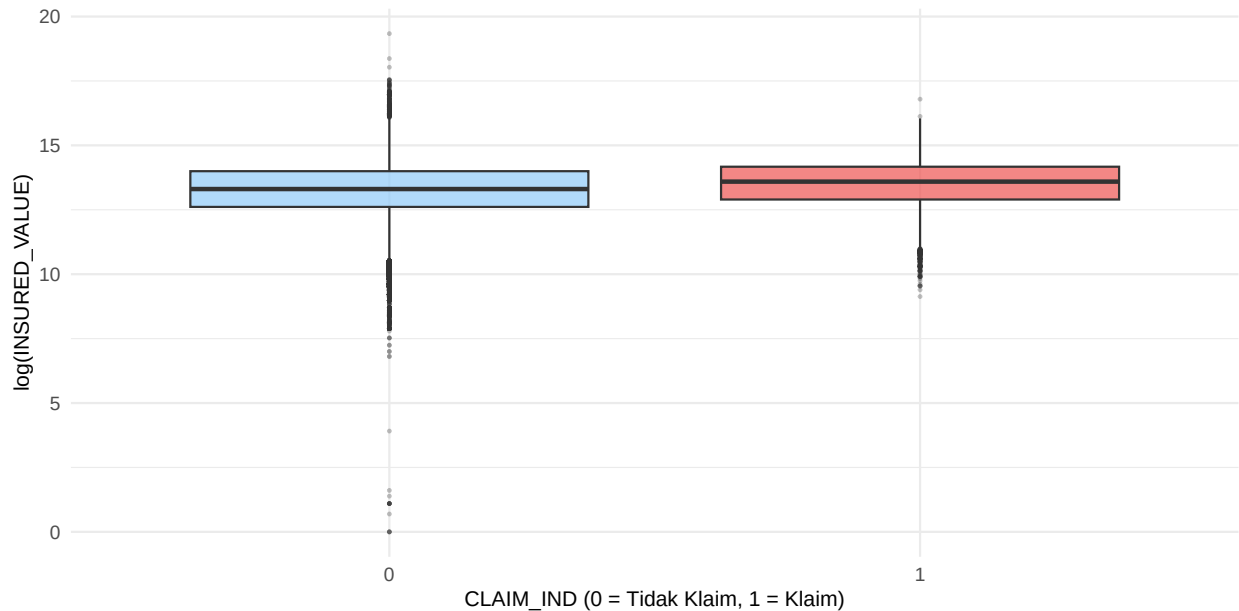
5.1 Boxplot INSURED_VALUE vs CLAIM_IND

```
p_iv_box <- ggplot(motor %>% filter(INSURED_VALUE > 0),
  aes(x = CLAIM_IND, y = log(INSURED_VALUE), fill = CLAIM_IND)) +
  geom_boxplot(alpha = 0.7, outlier.size = 0.5, outlier.alpha = 0.3) +
  scale_fill_manual(values = c("0" = "#90CAF9", "1" = "#EF5350")) +
  labs(title = "Gambar 17. log(INSURED_VALUE) vs CLAIM_IND",
    subtitle = "Apakah nilai pertanggungan berbeda antara polis klaim dan tidak?",
    x = "CLAIM_IND (0 = Tidak Klaim, 1 = Klaim)",
    y = "log(INSURED_VALUE)") +
  theme(legend.position = "none")

p_iv_box
```

Gambar 17. log(INSURED_VALUE) vs CLAIM_IND

Apakah nilai pertanggungan berbeda antara polis klaim dan tidak?

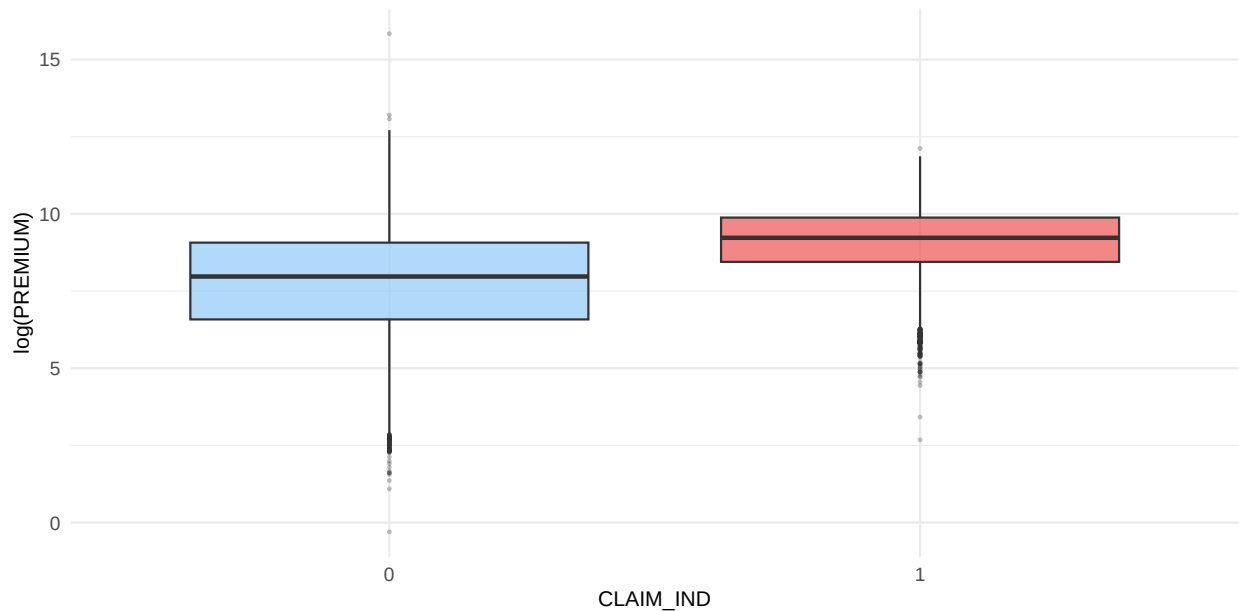


5.2 Boxplot PREMIUM vs CLAIM_IND

```
p_prem_box <- ggplot(motor %>% filter(PREMIUM > 0),  
  aes(x = CLAIM_IND, y = log(PREMIUM), fill = CLAIM_IND)) +  
  geom_boxplot(alpha = 0.7, outlier.size = 0.5, outlier.alpha = 0.3) +  
  scale_fill_manual(values = c("0" = "#90CAF9", "1" = "#EF5350")) +  
  labs(title = "Gambar 18. log(PREMIUM) vs CLAIM_IND",  
    subtitle = "Polis dengan premi lebih tinggi cenderung lebih banyak klaim?",  
    x = "CLAIM_IND", y = "log(PREMIUM)") +  
  theme(legend.position = "none")  
  
p_prem_box
```

Gambar 18. log(PREMIUM) vs CLAIM_IND

Polis dengan premi lebih tinggi cenderung lebih banyak klaim?

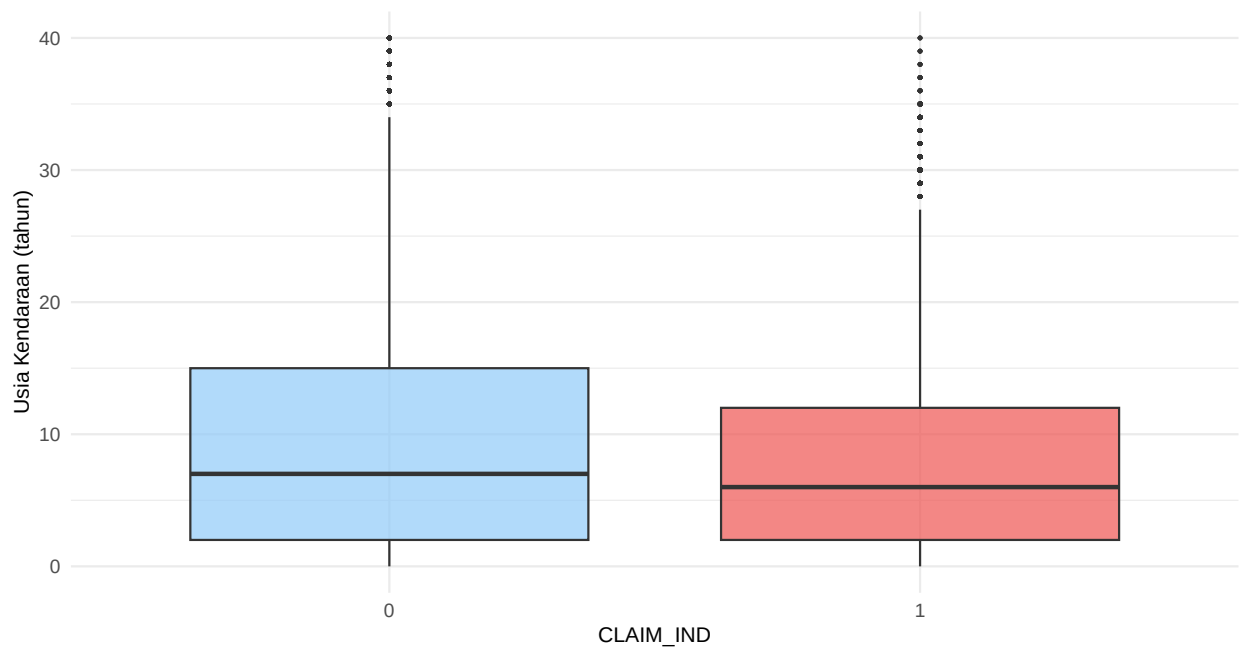


5.3 Boxplot VEHICLE_AGE vs CLAIM_IND

```
p_vage_box <- ggplot(motor %>% filter(VEHICLE_AGE >= 0 & VEHICLE_AGE <= 40),  
  aes(x = CLAIM_IND, y = VEHICLE_AGE, fill = CLAIM_IND)) +  
  geom_boxplot(alpha = 0.7, outlier.size = 0.5, outlier.alpha = 0.3) +  
  scale_fill_manual(values = c("0" = "#90CAF9", "1" = "#EF5350")) +  
  labs(title = "Gambar 19. Usia Kendaraan vs CLAIM_IND",  
    x = "CLAIM_IND", y = "Usia Kendaraan (tahun)") +  
  theme(legend.position = "none")
```

p_vage_box

Gambar 19. Usia Kendaraan vs CLAIM_IND



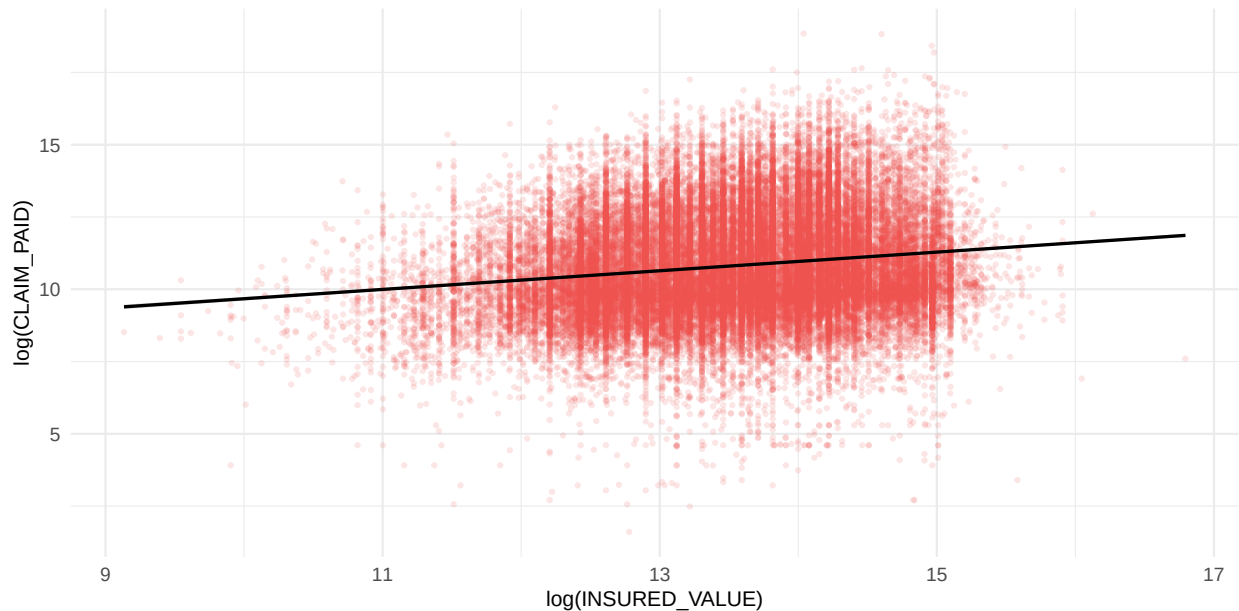
5.4 Scatter Plot $\log(\text{CLAIM_PAID})$ vs $\log(\text{INSURED_VALUE})$ — Subset Klaim

```
p_scatter <- ggplot(klaim_df %>% filter(INSURED_VALUE > 0),  
                    aes(x = log(INSURED_VALUE), y = log(CLAIM_PAID))) +  
  geom_point(alpha = 0.15, size = 0.8, colour = "#EF5350") +  
  geom_smooth(method = "lm", colour = "black", linewidth = 0.8, se = TRUE) +  
  labs(title = "Gambar 20. log(CLAIM_PAID) vs log(INSURED_VALUE) - Polis Berklaim",  
       subtitle = "Garis biru = regresi linier sederhana; shade = CI 95%",  
       x = "log(INSURED_VALUE)", y = "log(CLAIM_PAID)")
```

p_scatter

Gambar 20. log(CLAIM_PAID) vs log(INSURED_VALUE) — Polis Berklaim

Garis biru = regresi linier sederhana; shade = CI 95%



5.5 Violin Plot log(CLAIM_PAID) per TYPE_VEHICLE — Subset Klaim

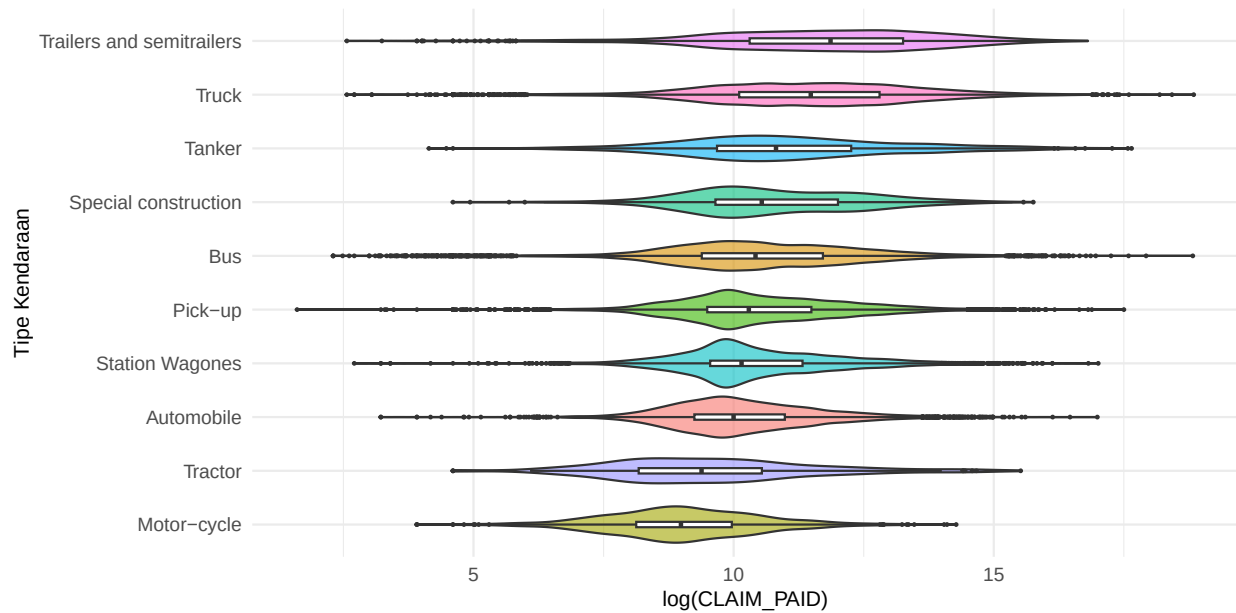
```
# Hanya tampilkan tipe kendaraan dengan lebih dari sama dengan 100 klaim
valid_types <- klaim_df %>%
  count(TYPE_VEHICLE) %>%
  filter(n >= 100) %>%
  pull(TYPE_VEHICLE)

p_violin <- ggplot(klaim_df %>% filter(TYPE_VEHICLE %in% valid_types),
  aes(x = fct_reorder(TYPE_VEHICLE, CLAIM_PAID, median),
    y = log(CLAIM_PAID),
    fill = TYPE_VEHICLE)) +
  geom_violin(alpha = 0.6, show.legend = FALSE) +
  geom_boxplot(width = 0.1, fill = "white", outlier.size = 0.5, show.legend = FALSE) +
  coord_flip() +
  labs(title = "Gambar 21. Distribusi log(CLAIM_PAID) per Tipe Kendaraan",
    subtitle = "Violin + boxplot - hanya tipe kendaraan dengan lebih dari sama dengan 100 klaim",
    x = "Tipe Kendaraan", y = "log(CLAIM_PAID)")

p_violin
```

Gambar 21. Distribusi $\log(\text{CLAIM_PAID})$ per Tipe Kendaraan

Violin + boxplot — hanya tipe kendaraan dengan lebih dari sama dengan 100 klaim



BAGIAN 6 — MATRIKS KORELASI (VARIABEL NUMERIK)

```
num_for_corr <- motor %>%
  filter(INSURED_VALUE > 0, PREMIUM > 0, CCM_TON > 0) %>%
  transmute(
    log_INSURED_VALUE = log(INSURED_VALUE),
    log_PREMIUM        = log(PREMIUM),
    PROD_YEAR,
    VEHICLE_AGE,
    log_CCM_TON        = log(CCM_TON),
    DURATION_DAYS,
    EXPOSURE_YR
  ) %>%
  na.omit()

# Sample 5000 baris agar ggpairs tidak terlalu berat
set.seed(42)
sample_corr <- num_for_corr %>% sample_n(min(5000, nrow(.)))

# Heatmap korelasi Pearson
corr_mat <- cor(num_for_corr, use = "pairwise.complete.obs")

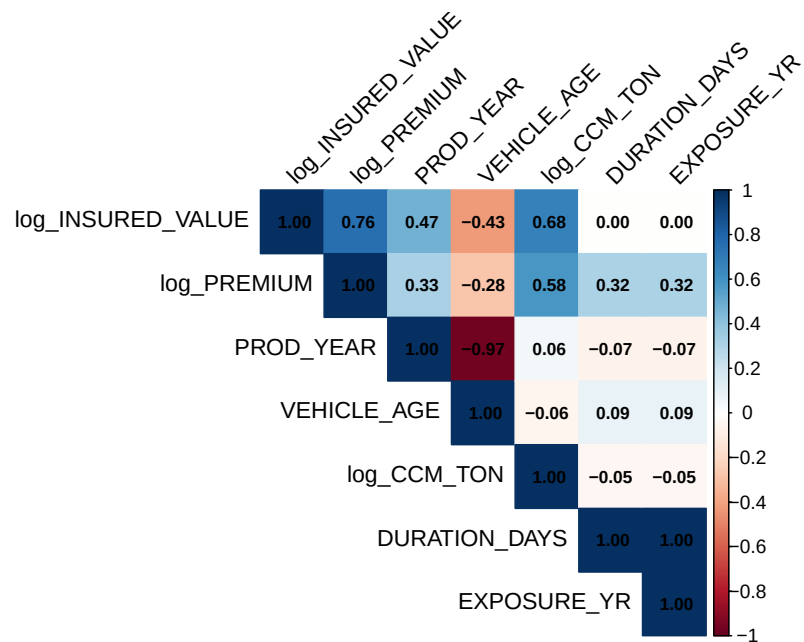
corrplot(corr_mat,
  method = "color",
  type   = "upper",
```

```

addCoef.col = "black",
number.cex  = 0.75,
tl.col      = "black",
tl.srt      = 45,
title       = "Gambar 22. Matriks Korelasi Variabel Numerik",
mar         = c(0, 0, 2, 0))

```

Gambar 22. Matriks Korelasi Variabel Numerik

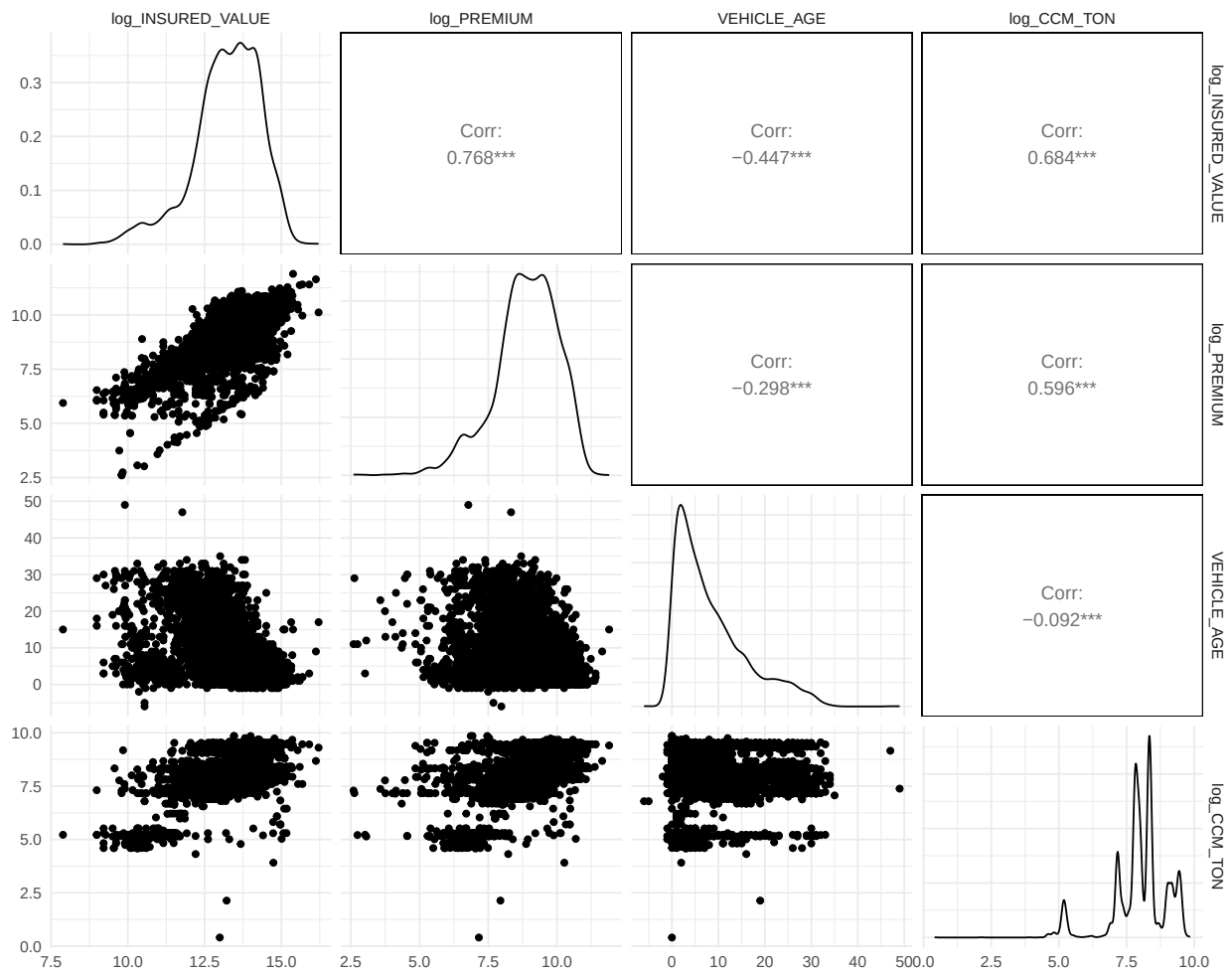


```

# Scatter matrix untuk subset variabel terpilih (tidak semua - agar terbaca)
ggpairs(
  sample_corr %>% select(log_INSURED_VALUE, log_PREMIUM, VEHICLE_AGE, log_CCM_TON),
  title = "Gambar 23. Scatter Matrix Variabel Numerik Terpilih (n=5.000 sampel)"
)

```

Gambar 23. Scatter Matrix Variabel Numerik Terpilih (n=5.000 sampel)



BAGIAN 7 — RINGKASAN TEMUAN EDA

```
cat("=====\n")

## =====
cat("RINGKASAN TEMUAN EDA - MOTOR INSURANCE DATA 2011-2018\n")

## RINGKASAN TEMUAN EDA - MOTOR INSURANCE DATA 2011-2018
cat("=====\n\n")

## =====
cat("1. CLAIM_IND (Variabel Respons Bab 7):\n")

## 1. CLAIM_IND (Variabel Respons Bab 7):
```

```

cat("    - Dataset sangat zero-inflated: ~7,5% polis mengajukan klaim\n")

##    - Dataset sangat zero-inflated: ~7,5% polis mengajukan klaim
cat("    - Justifikasi penggunaan GLM Logistik untuk Bab 7 sangat kuat\n\n")

##    - Justifikasi penggunaan GLM Logistik untuk Bab 7 sangat kuat
cat("2. CLAIM_PAID (Variabel Respons Bab 8):\n")

## 2. CLAIM_PAID (Variabel Respons Bab 8):
cat("    - Distribusi sangat right-skewed pada skala asli\n")

##    - Distribusi sangat right-skewed pada skala asli
cat("    - log(CLAIM_PAID) mendekati simetris => justifikasi log-link GLM Gamma\n\n")

##    - log(CLAIM_PAID) mendekati simetris => justifikasi log-link GLM Gamma
cat("3. Variabel Kategoris:\n")

## 3. Variabel Kategoris:
cat("    - SEX, INSR_TYPE, TYPE_VEHICLE, USAGE menunjukkan perbedaan claim rate\n")

##    - SEX, INSR_TYPE, TYPE_VEHICLE, USAGE menunjukkan perbedaan claim rate
cat("    - MAKE memiliki 792 level => perlu grouping atau fct_lump sebelum modeling\n\n")

##    - MAKE memiliki 792 level => perlu grouping atau fct_lump sebelum modeling
cat("4. Variabel Numerik:\n")

## 4. Variabel Numerik:
cat("    - INSURED_VALUE dan PREMIUM sangat right-skewed => transformasi log\n")

##    - INSURED_VALUE dan PREMIUM sangat right-skewed => transformasi log
cat("    - VEHICLE_AGE (turunan) menunjukkan pola claim rate yang menarik\n")

##    - VEHICLE_AGE (turunan) menunjukkan pola claim rate yang menarik
cat("    - PROD_YEAR dan log_CCM_TON berkorelasi tinggi dengan VEHICLE_AGE\n\n")

##    - PROD_YEAR dan log_CCM_TON berkorelasi tinggi dengan VEHICLE_AGE
cat("5. Tren Temporal:\n")

## 5. Tren Temporal:
cat("    - Portofolio tumbuh dari 2011 ke puncak, kemudian stabilisasi\n")

##    - Portofolio tumbuh dari 2011 ke puncak, kemudian stabilisasi
cat("    - Claim rate relatif stabil antar tahun (perlu dikonfirmasi plot)\n\n")

##    - Claim rate relatif stabil antar tahun (perlu dikonfirmasi plot)
cat("6. Implikasi Pemodelan:\n")

## 6. Implikasi Pemodelan:

```

```

cat("    - Bab 7 (Logistik): prediktor utama kandidat => SEX, INSR_TYPE,\n")

##    - Bab 7 (Logistik): prediktor utama kandidat => SEX, INSR_TYPE,
cat("        TYPE_VEHICLE, USAGE, log_INSURED_VALUE, log_PREMIUM, VEHICLE_AGE\n")

##        TYPE_VEHICLE, USAGE, log_INSURED_VALUE, log_PREMIUM, VEHICLE_AGE
cat("    - Bab 8 (Gamma/Tweedie): respons log(CLAIM_PAID), subset polis berklaim\n")

##    - Bab 8 (Gamma/Tweedie): respons log(CLAIM_PAID), subset polis berklaim
cat("    - Bab 6 (Poisson/NB): perlu membuat variabel CLAIM_COUNT per polis\n")

##    - Bab 6 (Poisson/NB): perlu membuat variabel CLAIM_COUNT per polis

```